TRAN QUANG DUYEN

AI Engineer · RAG · AI Agent · LLM Systems

tranquangduyen2503@gmail.com | +84 398 116 991 | github.com/duyentq-aie | duyentq-aie.github.io | Vinh Long, Vietnam

SUMMARY

AI Engineer with a solid Backend foundation (FastAPI, Microservices) gained through commercial experience delivering software for the Australian market. Proven ability to architect end-to-end RAG and Agentic workflows (LangGraph) with code-level guardrails, pre-deployment evaluation (Golden Sets, MRR/Recall@k), and per-request latency/cost observability (Langfuse). IT Graduate from FPT University ready to build reliable, scalable, and cost-efficient LLM applications.

EDUCATION & LANGUAGES

FPT University · Information Technology 2022 – 2026

English — technical reading; follows English-language work meetings.

SKILLS

AI / LLM: Python, ML/DL fundamentals, RAG, Embedding, Prompt engineering, AI Agent (LangChain/LangGraph), Vector Database, Fine-tuning

Evaluation & monitoring: Golden set, MRR / Recall@k, Latency / cost tracking, Tracing, Langfuse

Backend & data: FastAPI, REST API, Pydantic, Microservices, PostgreSQL, MongoDB

DevOps / MLOps: Docker, CI/CD (GitHub Actions), AWS (basic), Git, Linux (basic)

PROJECTS

ShopMate — AI sales assistant for a fashion store 10/2026

Personal project · github.com/duyentq-aie/shopmate

Python · LangGraph · Gemini · Qdrant · bge-m3 · Saleor (GraphQL) · FastAPI · Langfuse · GitHub Actions

- Built a shopping agent that answers product, return-policy and order-status questions on a Saleor store (250 products); read-only access, max 8 steps, support tickets only after the customer confirms.
- Raised search accuracy for Vietnamese queries typed without accents by **+0.43 MRR** (95% CI 0.27–0.58) by fusing accent-folded keyword search with bge-m3 semantic vectors in Qdrant.
- On a locked test set run once (33 questions): **31/33** correct answers, intent routing **33/33**, 2.6 s median latency, ~US$0.9 per 1,000 conversations; a code-level guardrail blocks replies with a wrong price.
- Evaluated like an experiment: 104-question golden set, test split locked and run only once; GitHub Actions CI runs 1,600+ automated tests and secret scanning.
- Shipped an embeddable chat widget, a request-tracing dashboard and opt-in Langfuse tracing with personal data masked.

DevPrep — AI mock-interview platform 2026

Capstone project (team) · github.com/Usopp204/DevPrep · FPT University

.NET 8 · ASP.NET Core · YARP · SQL Server · FastAPI · Gemini · ChromaDB · InsightFace · SpeechBrain · Deepgram · Polly · Docker

- Owned 3 of 10+ .NET 8 microservices (Interview, Evaluation, ASP.NET Core MVC web client) and co-owned the FastAPI AI service, behind a YARP gateway with per-service SQL Server databases, all in Docker Compose.
- Built the interview flow: questions picked by role, level and focus areas from a **34,678**-question bank; voice room with Deepgram speech-to-text, Piper text-to-speech, Gemini hints, skip, pause/resume and idle auto-advance.
- Built RAG-based grading: Gemini scores each answer against reference answers retrieved from ChromaDB (bge-m3) on a **50/30/20** rubric (knowledge / problem-solving / communication) based on O*NET data; background grading with recovery and a duplicate-grading guard.
- Built biometric anti-cheating with InsightFace (ArcFace) face and SpeechBrain (ECAPA-TDNN) voice embeddings: a verification gate before each session (3 attempts, no credit charged until it passes) and per-question face checks that flag no-face or multiple faces.
- Made credit charging race-free (deduction and session start in **one atomic conditional UPDATE**) and the enrollment bonus idempotent with a unique filtered index.
- Hardened the services: Polly circuit breakers on all outbound HTTP calls, per-route gateway timeouts, IDOR checks on reports, and 30 raw exception leaks replaced with clean 503 responses.

WORK EXPERIENCE

Backend Developer Intern 10/2024 – 02/2025

BIWOCO SOFTWARE COMPANY LIMITED · Software for the Australian market · 4 months

Python · FastAPI · MongoDB · AWS · Microservices · REST API

- Developed and maintained asynchronous RESTful APIs with FastAPI, using Pydantic schema validation, custom error handling and a clear separation of business logic.
- Implemented data persistence and query logic in MongoDB, including server-side pagination, multi-field filtering and field projection for CRUD endpoints.
- Contributed to feature development and bug fixing in a microservice-based backend deployed on AWS, keeping OpenAPI/Swagger docs up to date for frontend and inter-service integration.
- Attended English-language Agile stand-ups and sprint reviews with the Australian client team, following API specifications, edge cases and delivery milestones.

AWARDS & CERTIFICATES

- Top 15 — FPTU Can Tho Olympic AI 2026, "AI for Smart Mekong Delta" · FPT University Can Tho, 2026
- Practical AI Training Program · FPT University Can Tho, 07–08/2026